The Nonparametric Identification of Treatment Effects in Duration Models

Abbring & Van den Berg (2003, *Econometrica*)

Econometrics Reading Group

The question

Running example. An unemployed worker may receive a benefit sanction at a random time S during the spell. Outcome: unemployment duration Y.

Target. The causal effect of the sanction on the *exit rate* to work.

Why this is hard

1. **Dynamic treatment:** no fixed treated group; everyone is untreated until S.
2. **Treatment is censored:** if you find a job before your sanction date, S is never seen.
3. **Selection on unobservables:** low-motivation workers get sanctioned earlier *and* exit more slowly.
4. **No credible instrument:** whatever shifts sanctions, the worker likely sees and reacts to.

Punchline. The *timing* of S relative to Y does the job of the instrument, at the price of structure on hazards.

Primer 1: hazards and dynamic selection

Hazard of a duration T:

$$\theta(t) = \lim_{dt \downarrow 0} \frac{\Pr(T \in [t, t+dt) \mid T \geq t)}{dt}$$

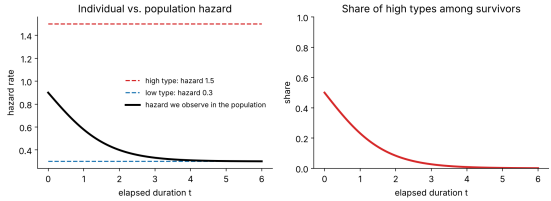
$\Lambda(t) = \int_0^t \theta, \quad \Pr(T > t) = e^{-\Lambda(t)}.$

In every instant, each survivor flips a coin with success probability $\theta(t)\,dt$.

Heterogeneity: $\theta(t \mid V) = \lambda(t)\,V$. Then the hazard we *observe* is

$$\bar{\theta}(t) = \lambda(t)\,\mathbb{E}[V \mid T \geq t],$$

and $\mathbb{E}[V \mid T \geq t]$ **falls**: high-V people leave first.



Trap or sorting? A falling exit rate can be true duration dependence ($\lambda \downarrow$) or composition ($\mathbb{E}[V \mid T \geq t] \downarrow$). This is Heckman's *true vs. spurious state dependence*, in continuous time.

Primer 2: the MPH model and the Laplace-transform toolkit

Mixed proportional hazard: $\theta(t \mid x, V) = \lambda(t)\,\varphi(x)\,V, \quad V \sim G, \quad V \perp x$. Then

$$\Pr(T > t \mid x) = \mathbb{E}\big[e^{-V\varphi(x)\Lambda(t)}\big] = \mathcal{L}\big(\varphi(x)\Lambda(t)\big), \qquad \mathcal{L}(a) := \mathbb{E}[e^{-aV}].$$

Toolkit. e^{-aV} is the probability that type V survives to index a. So derivatives of $\log \mathcal{L}$ are *moments among survivors*. In two dimensions, $\mathcal{L}(a, b) = \mathbb{E}[e^{-aV_Y - bV_S}]$:

$$-\partial_a \log \mathcal{L} = \mathbb{E}[V_Y \mid \text{surv.}], \qquad -\partial_b \log \mathcal{L} = \mathbb{E}[V_S \mid \text{surv.}], \qquad \partial^2_{ab} \log \mathcal{L} = \mathrm{Cov}(V_Y, V_S \mid \text{surv.}).$$

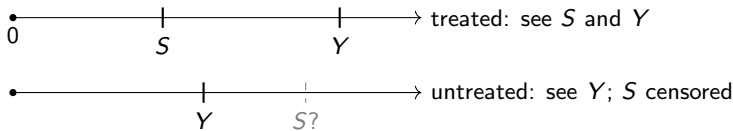
Single-risk identification (Elbers & Ridder 1982), in 3 steps

1. $t \downarrow 0$: nobody has been selected out yet, so hazard ratios across x give $\varphi(x)$ (needs $\mathbb{E}V < \infty$).
2. At t^* with $\Lambda(t^*) = 1$: $\Pr(T > t^* \mid x) = \mathcal{L}(\varphi(x))$. Varying x traces $\mathcal{L}$, hence G.
3. Any t: solve $\Pr(T > t \mid x) = \mathcal{L}(\varphi(x)\Lambda(t))$ for $\Lambda(t)$.

We will run exactly this logic in two dimensions.

Set-up: potential outcomes indexed by treatment time

- Treatments are dates $s \in [0, \infty]$ ($s = \infty$: never). Potential outcomes $Y^*(s)$; observed $Y = Y^*(S)$.
- **No anticipation:** $\theta_{Y^*(s)}(t) = \theta_{Y^*(s')}(t)$ for $t \leq \min(s, s')$. A future treatment date does not affect today's hazard.
  Reacting to the sanction *regime* is allowed (it is in every $Y^*(s)$), so the effect is *"treated at s vs. later, under this regime"*.



What the data reveal (all conditional on x):

- $P_0(t) = \Pr(Y > t, S > t)$: nothing has happened by t;
- $h_Y(t)$, $h_S(t)$: rate of exiting untreated / of being treated, among those with nothing yet;
- $P_1(y \mid s) = \Pr(Y > y \mid S = s,\ Y > s)$: post-treatment survival of the **cohort treated at** s.

Fact (Prop. 1): any such data can be rationalized with no anticipation *and* random assignment. So no anticipation is untestable, and we need a model of selection.

The model

Assumption 2 (selection on (x, V)): $Y^*(\cdot) \perp S \mid (x, V)$, with V time-invariant.

Model 1a.

$$\theta_Y(t \mid s, x, V) = \lambda_Y(t)\,\varphi_Y(x)\,V_Y \cdot \delta(t \mid s, x)^{\mathbf{1}\{t>s\}}, \qquad \theta_S(t \mid x, V) = \lambda_S(t)\,\varphi_S(x)\,V_S,$$

with $(V_Y, V_S) \sim G$ **arbitrarily dependent**, independent of x.

- Two MPH hazards. Their link through $\mathrm{Cov}(V_Y, V_S)$ is the **selection on unobservables**.
- δ is the **causal effect**: a hazard multiplier after treatment. It may vary with t, s (hence with $t - s$) and x, but not with V.
- No anticipation is built in: δ only acts for $t > s$.
- The same x enters both hazards, so there is **no exclusion restriction**. λ's, φ's, G, δ are all nonparametric.

Assn. 3: $\{(\varphi_Y(x), \varphi_S(x))\}$ contains an open set of $\mathbb{R}^2$. **Assn. 4:** $\mathbb{E}V_Y, \mathbb{E}V_S < \infty$.
Normalizations: $\varphi(x^*) = 1$, $\Lambda(t^*) = 1$.

The selection problem at the moment of treatment

Fix a duration s. Write $a_s = \varphi_Y(x)\Lambda_Y(s)$, $b_s = \varphi_S(x)\Lambda_S(s)$.

Who is still at risk (nothing happened by s): $dG_{R_s}(v) \propto e^{-a_s v_Y - b_s v_S}\, dG(v)$.

Who is treated at s: the risk set, reweighted by the treatment intensity, $dG_s(v) \propto v_S\, dG_{R_s}(v)$.

Exit hazards right at s:

$$\underbrace{\lambda_Y(s)\varphi_Y(x)\,\mathbb{E}_{R_s}[V_Y]}_{\text{not yet treated}} \qquad \text{vs.} \qquad \underbrace{\lambda_Y(s)\varphi_Y(x)\,\delta(s \mid s,x)\,\mathbb{E}_s[V_Y]}_{\text{just treated}}$$

$$\frac{\text{just treated}}{\text{not yet treated}} = \delta(s \mid s,x) \times \underbrace{\left(1 + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s}[V_Y]\,\mathbb{E}_{R_s}[V_S]}\right)}_{\text{selection factor}}$$

- If $V_Y \perp V_S$ among survivors, timing is as good as random at s: the ratio *is* the causal effect.
- Otherwise we need the **joint distribution of types among survivors**. The pre-treatment data will supply it.

Timing as an instrument

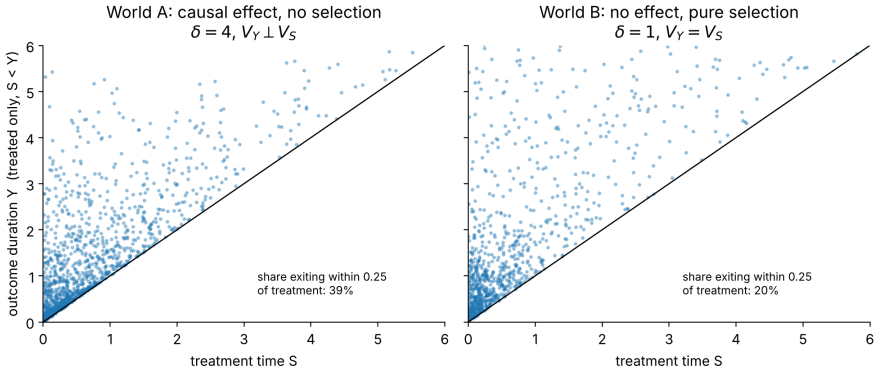
	Static selection / IV	Timing of events
Exogenous variation in treatment	excluded variable Z	the timing shock: given (x, V), $S = \Lambda_S^{-1}(\frac{E_S}{\varphi_S(x)V_S})$, $E_S \sim \text{Exp}(1)$
What is excluded from the outcome	the *variable* Z	the *shock* E_S (via Assn. 2)
Correcting for who selects in	control function $\mathbb{E}[U \mid D = 1, Z]$	selection factor / G_s, from the competing-risks data
Covariate requirement	Z shifts D, not Y	x shifts both hazards, *non-proportionally* (Assn. 3)

Why timing is informative. The treatment acts only *after* S; V acts *throughout*.

- Causal effect $\Rightarrow$ **local** dependence: exits bunch right after S, *whenever* S occurs.
- Selection $\Rightarrow$ **global** dependence: early-treated people are early leavers in general.

In a static latent-index model the treatment effect needs an exclusion restriction. Here the timing replaces it.

What the data look like



Treated observations only. Left: $\delta = 4$, independent frailties. Right: $\delta = 1$, $V_Y = V_S$. A causal effect makes points hug the diagonal *at every* S. Selection produces quick exits only where high types are concentrated (small S).

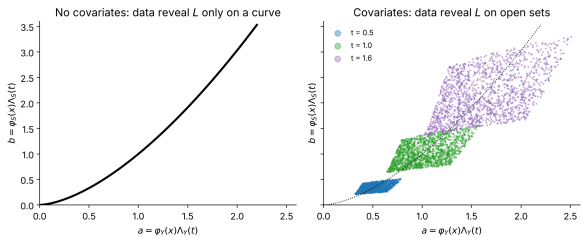
The pre-treatment data are a competing-risks problem

Before treatment there are two latent clocks: Y^0 (exit if never treated) and S. Their joint survival is

$$\Pr(Y^0 > y,\ S > s \mid x) = \mathcal{L}\big(\varphi_Y(x)\Lambda_Y(y),\ \varphi_S(x)\Lambda_S(s)\big). \qquad (\delta \text{ does not appear!})$$

We only see the *first* event: the diagonal $y = s = t$, i.e. $P_0(t \mid x) = \mathcal{L}(a_t(x), b_t(x))$, plus which event it was.

Tsiatis (1975): without structure, any such data are generated by *independent* clocks. The dependence, i.e. the selection, is unidentified.



Without x, the data reveal $\mathcal{L}$ only along a curve $t \mapsto (\Lambda_Y(t), \Lambda_S(t))$. A curve cannot pin down a function of two variables. With x shifting the two hazards separately, the data reveal $\mathcal{L}$ on *open sets*.

Identifying the competing-risks part (Prop. 2): three steps

Step 1: φ's from short durations (nobody has been selected out yet):

$$\frac{h_Y(t \mid x)}{h_Y(t \mid x^*)} = \frac{\varphi_Y(x)\,\mathbb{E}[V_Y \mid R_t, x]}{\mathbb{E}[V_Y \mid R_t, x^*]} \xrightarrow{t\downarrow 0} \varphi_Y(x), \qquad \text{same for } \varphi_S.$$

Uses *which* event happened and $\mathbb{E}V < \infty$.

Step 2: G at the normalization date ($\Lambda_Y(t^*) = \Lambda_S(t^*) = 1$):

$$P_0(t^* \mid x) = \mathcal{L}\big(\varphi_Y(x), \varphi_S(x)\big).$$

Known points fill an open set (Assn. 3), so $\mathcal{L}$ is known there. $\mathcal{L}$ is real-analytic, so it is known on $(0,\infty)^2$, and $\mathcal{L}$ determines G, **including the dependence between V_Y and V_S**.

Step 3: baselines. For every t, $(\Lambda_Y(t), \Lambda_S(t))$ is the unique (a, b) with

$$P_0(t \mid x) = \mathcal{L}\big(\varphi_Y(x)\,a,\ \varphi_S(x)\,b\big) \quad \text{for all } x.$$

$\Rightarrow \lambda_Y, \lambda_S, \varphi_Y, \varphi_S, G$ are identified **without touching post-treatment data**.

Where the selection shows up in the data

From the toolkit, applied to the observed $P_0(t \mid x) = \mathcal{L}(a_t(x), b_t(x))$:

$$\frac{\partial^2 \log P_0}{\partial a \, \partial b} = \mathrm{Cov}\left(V_Y, V_S \mid \text{nothing happened by } t,\, x\right).$$

- $V_Y \perp V_S \iff \mathcal{L}(a,b) = \mathcal{L}_Y(a)\mathcal{L}_S(b) \iff \log P_0$ is **additively separable** in the two covariate indices.
- Selection on unobservables = an **interaction**: a covariate that raises the treatment hazard changes how the exit index affects survival.
- Seeing an interaction requires the two indices to move separately (Assn. 3). That is where the instrument-like requirement hides: *no exclusion, but non-proportional covariate effects*.

The selection factor of slide 7 is then a function of identified objects:

$$1 + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s} V_Y \; \mathbb{E}_{R_s} V_S} = \frac{\mathcal{L}\,\mathcal{L}_{ab}}{\mathcal{L}_a\,\mathcal{L}_b}\Big|_{(a_s, b_s)} \quad \Rightarrow \quad \delta(s \mid s, x) = \frac{\text{observed hazard ratio at } s}{\mathcal{L}\mathcal{L}_{ab}/(\mathcal{L}_a\mathcal{L}_b)}.$$

Stripping out selection after treatment (Prop. 3)

For $y > s$, with $D(y \mid s, x) = \int_s^y \lambda_Y(\tau)\delta(\tau \mid s, x)\, d\tau$ the integrated post-treatment hazard,

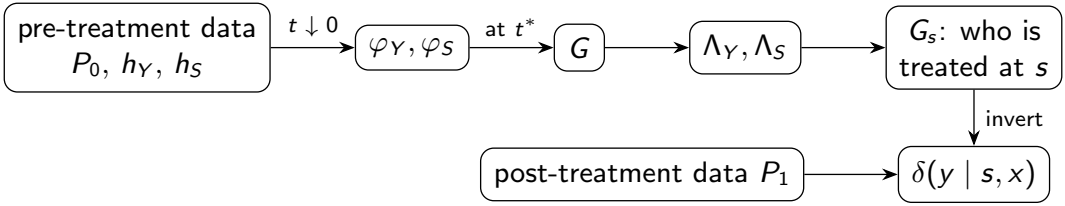
$$P_1(y \mid s, x) = \mathbb{E}_{G_s}\Big[e^{-\varphi_Y(x) V_Y\, D(y \mid s, x)}\Big] = \frac{\mathcal{L}_b\big(a_s + \varphi_Y(x) D(y \mid s, x),\ b_s\big)}{\mathcal{L}_b(a_s, b_s)}.$$

- The left side is data. On the right, everything except D is known from Prop. 2.
- The right side is strictly decreasing in D, so **invert** to get $D(y \mid s, x)$. Then $\delta(y \mid s, x) = \partial_y D / \lambda_Y(y)$.

Two selection problems, both solved by knowing G:

1. **Selection into treatment at s:** the V_S-tilt of G_s (absent if $V_Y \perp V_S$).
2. **Selection out after s:** the treated cohort's survivors drift toward low V_Y. This happens whenever V_Y varies, *even with random timing*. So hazard ratios at longer horizons $\neq \delta$.

The identification chain, and where each assumption is used



Assumption	Role
No anticipation	pre-treatment period is a clean competing-risks problem (δ absent)
Selection only via (x, V)	clocks independent given V; treated = risk set reweighted by V_S
MPH separability, $V \perp x$	index structure $\mathcal{L}(\varphi\Lambda)$ behind all three steps
$\mathbb{E}V < \infty$	step 1 (short-duration hazard ratios)
2-D covariate variation	step 2: trace $\mathcal{L}$ on an open set $\Rightarrow$ identifies the dependence

Heterogeneous effects (Model 1b, Prop. 4)

$$\theta_Y(t \mid s, x, V) = \begin{cases} \lambda_Y(t)\varphi_Y(x)V_Y & t \le s \\ \lambda_\Delta(t)\varphi_\Delta(x)V_\Delta & t > s \end{cases}, \qquad (V_Y, V_\Delta, V_S) \sim G_\Delta,$$

$$\text{so the effect is} \quad \delta(t \mid x, V) = \frac{\lambda_\Delta(t)\varphi_\Delta(x)V_\Delta}{\lambda_Y(t)\varphi_Y(x)V_Y}.$$

- **Gained:** the effect varies with unobservables, and V_Δ may correlate with V_S. This allows **selection on gains** (Roy-type sorting).
- **Lost:** δ can no longer depend on the treatment date s.
- **Why it works:** after S, the residual duration of the treated cohort is again MPH, in $\Lambda_\Delta(y) - \Lambda_\Delta(s)$ with frailty V_Δ, so Elbers–Ridder can be rerun on it.
- **Extra assumptions:** variation in $\varphi_\Delta(x)$, and $\mathbb{E}[V_\Delta V_S] < \infty$. The moment is of the *product* because the treated cohort is tilted by V_S.

Multiple spells: the within estimator (Model 2c, Prop. 6)

Strata (same person twice, siblings, . . .) with two spells sharing the same unobservables:

$$\theta_{Y_k}(t \mid S_k, V) = \lambda_{Y_k}(t)\, V_Y(t)\, \delta_k(t \mid S_k)^{\mathbf{1}\{t > S_k\}}, \qquad k = 1, 2, \quad \lambda_{Y_1} \equiv 1.$$

$V_Y(t)$ may be **any process common to both spells**, and there is **no model for** S_k.

Given that the first exit in the stratum happens at t, which spell was it? $V_Y(t)$ **cancels**:

$$\Pr(\text{spell } 1 \mid t, \text{histories}, V) = \frac{\theta_{Y_1}(t)}{\theta_{Y_1}(t) + \theta_{Y_2}(t)}.$$

treated at t	Pr(spell 1 first)	identifies
neither	$1/(1+\lambda_{Y_2})$	λ_{Y_2}
spell 1 only	$\delta_1/(\delta_1+\lambda_{Y_2})$	δ_1
spell 2 only	$1/(1+\lambda_{Y_2}\delta_2)$	δ_2
both	$\delta_1/(\delta_1+\lambda_{Y_2}\delta_2)$	(overidentifying)

Cox partial likelihood = conditional logit = fixed effects. Cost: V identical across spells, and **no causal links across spells**.

What is traded, and critiques

Not needed

- an excluded instrument
- selection on observables
- parametric functional forms
- multiple spells (single-spell results)

Needed

- no anticipation (untestable)
- time-invariant unobservables, $V \perp x$
- MPH proportionality
- 2-D, non-proportional covariate variation
- finite means; continuous time

Critiques

- Identification comes from $t \downarrow 0$, where data are thin. In practice: parametric λ, discrete G (Heckman–Singer); Monte Carlo evidence in Gaure, Røed & Zhang (2007).
- Discrete time: sanction and exit in the same month blur the local signal.
- δ is a hazard ratio *conditional on* V. Effects on mean duration or population survival require G.
- Transitory shocks hitting both hazards (the caseworker reacts to this week's low effort) violate Assn. 2.

Connection: staggered adoption DiD

Same skeleton

- $Y^*(s) \leftrightarrow Y_t(g)$, potential outcomes indexed by adoption cohort.
- No anticipation is *the same assumption* (Callaway & Sant'Anna 2021; Sun & Abraham 2021; Borusyak, Jaravel & Spiess 2024).
- The risk set $R_s \leftrightarrow$ the *not-yet-treated* comparison group.
- $\delta(t \mid s)$ varies by cohort s and event time $t - s$: the heterogeneity that breaks TWFE.

Differences

- Selection: *parallel trends* (additive, in levels) vs. *time-invariant frailty* (multiplicative, in hazards) plus a random timing shock.
- **Attrition from an absorbing exit:** the treatment changes who remains (slide 13), so "parallel trends in hazards" fails *even under random timing*.
  DiD on survival probabilities $\rightarrow$ a population estimand; ToE $\rightarrow$ the individual hazard effect δ, which needs G.
- If timing is random given observables (*dynamic CIA*), treat later treatment as independent censoring. Population effects on survival are then identified nonparametrically (Fredriksson & Johansson 2008; Crépon et al. 2009; Vikström 2017). ToE is the tool when timing is selected on *unobservables*.

Connection: sequential exogeneity, feedback, and traps

Sequential exogeneity and feedback

- A hazard is an intensity relative to the past: $dN_Y(t) - \theta_Y(t \mid \text{past}, V)\,dt$ is a martingale difference. This is a continuous-time *sequential moment condition*. No anticipation means treatment is *predetermined*.
- In a single absorbing spell, outcome→treatment feedback is degenerate (the only outcome history is "still here"). The bite is on transitory shocks: S may depend on permanent V, not on shocks to the exit rate. This is *strict exogeneity given a fixed effect*.
- Model 2c moves toward sequential exogeneity: S is unmodelled and may track a time-varying common $V_Y(t)$, at the cost of needing multiple spells.

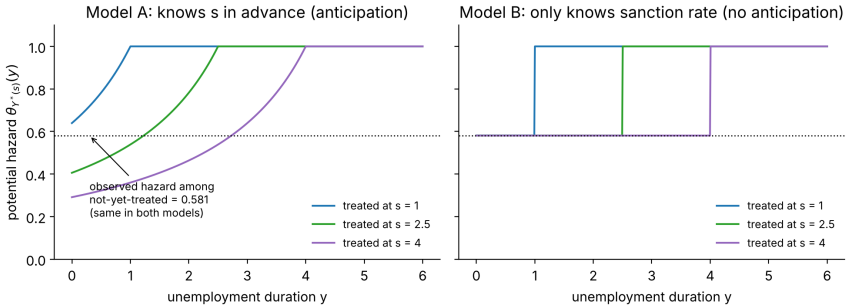
Traps: state dependence vs. permanent heterogeneity

- The model separates three objects: $\lambda_Y \downarrow$ (true duration dependence: a trap), G (permanent heterogeneity), and δ (the effect of an intervention).
- Traps suggest *when* you intervene matters: $\delta(t \mid s)$ varying in s. Model 1a allows it; Model 1b rules it out.
- Multiple-spell results assume no cross-spell effects, i.e. they *rule out* lagged duration dependence, the across-spell trap mechanism.

Takeaways

1. **Timing is information.** Treatment acts only after S, while unobservables act throughout. Causal effects are *local*; selection is *global*.
2. **The pre-treatment period is a competing-risks experiment with no treatment effect in it.** With MPH structure and non-proportional covariate variation, it reveals who is at risk and who gets treated (G). Knowing G strips out selection into treatment and selection out of the risk set afterwards.
3. **The price.** No anticipation, multiplicative separability, time-invariant heterogeneity, and a timing shock that is excluded from the outcome. Multiple spells buy back generality at the cost of no cross-spell dynamics.

Appendix A: non-identification (Prop. 1) and the job-search example



Model A: the worker knows the sanction date s, and search effort (hence the hazard) rises before s. Model B: only the sanction rate is known, so the hazard jumps at s. With suitably chosen parameters the two produce *identical* data. The data only reveal the hazard among the not-yet-treated (0.581 in both).

Proof of Prop. 1. By Tsiatis/Miller there is an independent pair $(\hat{Y}, \hat{S})$ matching the competing-risks part. Use $\hat{S}$ as the treatment and $\hat{Y}$'s hazard before treatment for every $Y^*(s)$; define post-treatment hazards from P_1.

Appendix B: data objects vs. the paper's (Q_S, Q_Y)

The paper's data: $Q_S(y,s) = \Pr(Y > y, S > s, Y > S)$ and $Q_Y(y) = \Pr(Y > y, Y < S)$. Then

$$
\begin{aligned}
P_0(t) &= Q_S(t,t) + Q_Y(t), \\
h_Y(t) &= -Q_Y'(t)/P_0(t), \qquad h_S(t) = -\partial_s Q_S(t,s)\big|_{s=t}/P_0(t), \\
P_1(y \mid s) &= \partial_s Q_S(y,s)/\partial_s Q_S(s,s) \qquad (y > s).
\end{aligned}
$$

Conversely, (P_0, h_Y, h_S, P_1) determine (Q_S, Q_Y). The objects are equivalent.

Appendix C: uniqueness in step 3

Suppose $(a, b) \neq (a', b')$ both satisfy $P_0(t \mid x) = \mathcal{L}(\varphi_Y(x)a, \varphi_S(x)b)$ for all x. Then $\mathcal{L}(pa, qb) = \mathcal{L}(pa', qb')$ on an open set of (p, q). By analyticity,

$$\mathcal{L}(u, v) = \mathcal{L}(ku, mv) \quad \text{for all } (u, v) \in (0, \infty)^2, \qquad k = a'/a,\ m = b'/b,\ (k, m) \neq (1, 1).$$

- $k, m \geq 1$ (or both ≤ 1): contradicts $\mathcal{L}$ strictly decreasing in each argument (needs $\Pr(V \in (0, \infty)^2) > 0$).
- $k > 1 > m$: iterating gives $\mathcal{L}(u, v) = \mathcal{L}(k^n u, m^n v) \to \Pr(V_Y = 0)$ as $n \to \infty$. So $\mathcal{L}$ is constant, a contradiction. The case $k < 1 < m$ is symmetric.

Appendix D: proof sketches for Models 1b, 2a, 2b

Model 1b (Prop. 4). After Prop. 2, for $y > s$:

$$\frac{\partial^2 Q_S(y, s \mid x)/\partial y \partial s}{\partial^2 Q_S(y, s \mid x^*)/\partial y \partial s} \xrightarrow{y, s \downarrow 0} \varphi_S(x)\varphi_\Delta(x) \quad (\text{needs } \mathbb{E}[V_\Delta V_S] < \infty),$$

which identifies φ_Δ. Then $\partial_s Q_S/(\lambda_S \varphi_S) = \mathcal{L}^{(S)}_{G_\Delta}\big(\varphi_Y \Lambda_Y(s), \varphi_\Delta(\Lambda_\Delta(y) - \Lambda_\Delta(s)), \varphi_S \Lambda_S(s)\big)$. Let $s \downarrow 0$ to get Λ_Δ; vary x, y to trace $\mathcal{L}_{G_\Delta}$ on an open set.

Models 2a/2b (Prop. 5). Two spells per stratum sharing $V = (V_Y, V_S)$ (or (V_Y, V_Δ, V_S)), spell-specific baselines and effects. The two-spell competing-risks structure identifies $\Lambda_{Y_k}, \Lambda_{S_k}$ (Abbring & Van den Berg 2003, JRSSB). The extra cross-spell information then identifies δ_k and G, **without covariates and without finite means**, mirroring Honoré (1993).